



GATE ME 90 Days Challenge

Day 10/90 · Practice Set with Detailed Solutions

SUBJECT

Strength of Materials

TODAY'S TOPIC

Deflection of Beams

COUNTDOWN

100 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on the elastic curve equation, standard cases, superposition, Macaulay's method, moment-area theorems and propped cantilevers. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 (MCQ)

The differential equation of the elastic curve of a beam (small deflections, y upward, sagging moment positive) is

(A) $EI \frac{d^2y}{dx^2} = M$

(B) $EI \frac{d^2y}{dx^2} = V$

(C) $EI \frac{dy}{dx} = M$

(D) $EI \frac{d^4y}{dx^4} = M$

Q2 (NAT)

A simply supported beam of span 4 m carries a 20 kN point load at midspan. With $EI = 10000 \text{ kN}\cdot\text{m}^2$, the maximum deflection (in mm, round off to two decimal places) is _____

Q3 (MCQ)

The slope at each support of a simply supported beam of span L carrying a UDL w over the whole span is

(A) $wL^3/24EI$

(B) $wL^3/6EI$

(C) $wL^3/48EI$

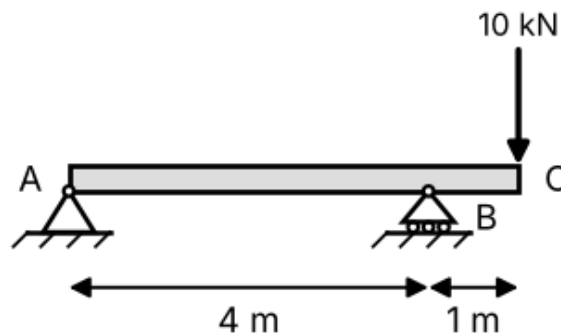
(D) $5wL^3/384EI$

Q9 MCQ

A propped cantilever of span L (fixed at A, propped at B) carries a point load P at midspan. The reaction at the prop is

(A) $5P/16$ **(B)** $11P/16$ **(C)** $P/2$ **(D)** $3P/8$ **Q10** NAT

The overhanging beam ABC shown (span $AB = 4$ m, overhang $BC = 1$ m, $EI = 10000$ kN·m²) carries a 10 kN load at the free end C. The downward deflection of C (in mm, round off to two decimal places) is _____

**Q11** MCQ

For a beam of span L carrying a UDL over the whole span, the ratio of the maximum deflection with both ends fixed to that with both ends simply supported is

(A) $1/5$ **(B)** $1/4$ **(C)** $1/2$ **(D)** $1/8$ **Q12** NAT

A cantilever of length L carries a point load at its free end. The deflection at the mid-length, as a fraction of the free-end deflection, is _____ (round off to four decimal places)

Q13 MCQ

A beam is bent by equal and opposite couples at its ends (pure bending). Its elastic curve is

(A) a circular arc**(B)** a parabola**(C)** a cubic curve**(D)** a straight line**Q14** NAT

A simply supported floor beam of span 8 m carries a UDL of 15 kN/m. With $E = 200$ GPa and a deflection limit of span/360, the minimum second moment of area required is _____ $\times 10^6$ mm⁴

Q15 MCQ

Two simply supported beams are identical except that the second has twice the span. Each carries the same central point load. Compared with the first, the second beam has

(A) 2 times the maximum bending stress and 8 times the maximum deflection**(B)** 2 times the stress and 4 times the deflection**(C)** 4 times the stress and 8 times the deflection**(D)** the same stress and 8 times the deflection

Answer Key

Q	Type	Answer
Q1	MCQ	(A) $EI \frac{d^2y}{dx^2} = M$
Q2	NAT	2.67
Q3	MCQ	(A) $wL^3/24EI$
Q4	NAT	7.25
Q5	MCQ	(A) one-eighth
Q6	NAT	5.11
Q7	MCQ	(A) the moment, about B, of the M/EI diagram area between A and B
Q8	NAT	4
Q9	MCQ	(A) $5P/16$
Q10	NAT	1.67
Q11	MCQ	(A) $1/5$
Q12	NAT	0.3125
Q13	MCQ	(A) a circular arc
Q14	NAT	180
Q15	MCQ	(A) 2 times the maximum bending stress and 8 times the maximum deflection

Detailed Solutions

Q1 Answer: (A) $EI \frac{d^2y}{dx^2} = M$

Concept: Curvature $1/R = M/(EI)$, and for small slopes $1/R \approx d^2y/dx^2$.

So $EI y'' = M$; differentiating, $EI y''' = V$ and $EI y'''' = -w$ (load intensity).

Exam tip: Integrate M once for slope, twice for deflection; fix the two constants from the supports.

Q2 Answer: 2.67

Concept: Central point load on a simply supported beam: $\delta_{\max} = PL^3/(48EI)$ at midspan.

$$\delta = 20 \times 4^3 / (48 \times 10000) = 1280 / 480000 = 0.002667 \text{ m} = \mathbf{2.67 \text{ mm}}$$

Exam tip: kN and $\text{kN}\cdot\text{m}^2$ give metres directly; multiply by 1000 for mm.

Q3 Answer: (A) $wL^3/24EI$

Concept: By symmetry the slope is zero at midspan; $\theta_A = \text{area of } M/EI \text{ diagram from A to midspan}$.

$$\text{Area of half the parabola} = (2/3) \times (L/2) \times (wL^2/8) = wL^3/24 \rightarrow \theta_A = \mathbf{wL^3/(24EI)}$$

Exam tip: Central point load gives $\theta_A = PL^2/(16EI)$.

Q4 Answer: 7.25

Concept: Superposition: add the deflections due to each load acting alone (linear elastic beam).

End load: $PL^3/(3EI) = 5 \times 27/(3 \times 9000) = 0.00500 \text{ m}$

UDL: $wL^4/(8EI) = 2 \times 81/(8 \times 9000) = 0.00225 \text{ m}$

$\delta_B = 5.00 + 2.25 = \mathbf{7.25 \text{ mm}}$

Exam tip: Memorise the 4 basic cantilever and 4 basic SS-beam cases; most GATE deflection questions are superpositions of them.

Q5 Answer: (A) one-eighth

Concept: $\delta \propto 1/I$ and $I = bd^3/12 \propto d^3$.

Doubling d multiplies I by 8 $\rightarrow$ deflection becomes **1/8**.

Bending stress ($\propto 1/d^2$) would fall to 1/4.

Exam tip: Depth is the cheapest way to stiffen a beam.

Q6 Answer: 5.11

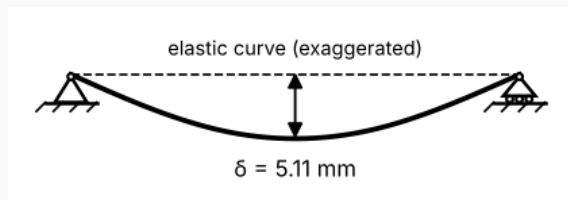
Concept (Macaulay): $R_A = R_B = 20 \text{ kN}$. $EI y'' = 20x - 20(x-2) - 20(x-4)$.

$EI y = (20/6)x^3 - (20/6)(x-2)^3 - (20/6)(x-4)^3 + C_1x$; $y(0) = 0$ gives $C_2 = 0$, and $y(6) = 0$ gives $C_1 = -80$.

At $x = 3$: $EI y = (20/6)(27) - (20/6)(1) - 80 \times 3 = 90 - 3.33 - 240 = -153.33 \text{ kN}\cdot\text{m}^3$

$\delta = 153.33/30000 = 0.00511 \text{ m} = \mathbf{5.11 \text{ mm}}$ (check: $Pa(3L^2 - 4a^2)/(24EI) = 20 \times 2 \times 92/(24 \times 30000) \checkmark$)

Exam tip: Terms in Macaulay brackets are dropped whenever the bracket is negative.

**Q7 Answer: (A) the moment, about B, of the M/EI diagram area between A and B**

Concept: $t_{B/A} = \int (M/EI) \bar{x}_B dx$ from A to B, with distances measured from **B** (the point whose deviation is wanted).

Taking moments about A instead gives $t_{A/B}$, the deviation of A from the tangent at B.

Exam tip: For a cantilever, take the tangent at the fixed end (it is horizontal), so $t_{B/A}$ is the deflection itself.

Q8 Answer: 4

Concept: An end couple gives a constant bending moment M_0 over the whole length.

$\theta = M_0L/EI = 10 \times 2/5000 = 0.004 \text{ rad}$

$\delta = M_0L^2/(2EI) = 10 \times 4/(2 \times 5000) = 0.004 \text{ m} = \mathbf{4 \text{ mm}}$

Exam tip: The elastic curve under constant M is a circular arc of radius $R = EI/M_0$.

Q9 Answer: (A) 5P/16

Concept: Compatibility: the prop deflection is zero.

Without the prop, the free end deflects $5PL^3/(48EI)$ (load at mid-length). The prop force R lifts it by $RL^3/(3EI)$.

$RL^3/3EI = 5PL^3/48EI \rightarrow R = 5P/16$; the fixed end takes $11P/16$.

Exam tip: $3wL/8$ is the prop reaction for a UDL. Do not mix the two.

Q10 Answer: 1.67

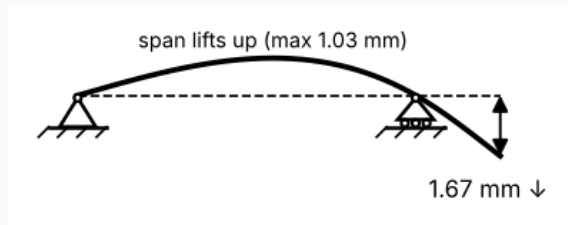
Concept: $\delta_C = (\text{deflection of BC as a cantilever}) + (\text{rotation of the span at B}) \times BC$.

The moment at B is $Pa = 10 \text{ kN}\cdot\text{m}$; it rotates the simply supported span at B by $\theta_B = (Pa)L/(3EI) = 10 \times 4/30000 = 0.001333 \text{ rad}$.

Cantilever part: $Pa^3/(3EI) = 10 \times 1/30000 = 0.000333 \text{ m}$; rotation part: $\theta_B \times a = 0.001333 \text{ m}$

$\delta_C = 0.000333 + 0.001333 = 0.001667 \text{ m} = 1.67 \text{ mm} \downarrow = Pa^2(L + a)/(3EI)$. The span AB bows upward (max 1.03 mm).

Exam tip: In overhang problems, never forget the rotation at the support carried to the tip.

**Q11 Answer: (A) 1/5**

Concept: Fixed-fixed: $wL^4/(384EI)$. Simply supported: $5wL^4/(384EI)$.

Ratio = $1/5$.

Exam tip: For a central point load the same ratio is $1/4$ ($PL^3/192EI$ vs $PL^3/48EI$).

Q12 Answer: 0.3125

Concept: For an end load, $y(x) = Px^2(3L - x)/(6EI)$, x from the fixed end.

$y(L/2) = P(L^2/4)(5L/2)/(6EI) = 5PL^3/(48EI)$; $y(L) = PL^3/(3EI)$

Ratio = $(5/48)/(1/3) = 5/16 = 0.3125$

Exam tip: By Maxwell's reciprocal theorem, $5PL^3/48EI$ is also the free-end deflection for a load at mid-length.

Q13 Answer: (A) a circular arc

Concept: $1/R = M/(EI)$; with M constant along the beam, the curvature is constant.

Constant curvature means a **circular arc**. The small-slope equation $y'' = M/EI$ gives a parabola, which is the same arc to first order.

Exam tip: A UDL gives a quartic curve and a point load a cubic between loads.

Q14 Answer: 180

Concept: Serviceability: $5wL^4/(384EI) \leq L/360$.

$$I \geq 5wL^3 \times 360/(384E) = 5 \times 15 \times 8^3 \times 360/(384 \times 200 \times 10^6) \text{ m}^4 \text{ (w in kN/m, E in kN/m}^2\text{)}$$

$$I \geq 1.8 \times 10^{-4} \text{ m}^4 = 1.8 \times 10^8 \text{ mm}^4 = \mathbf{180 \times 10^6 \text{ mm}^4}$$

Exam tip: $1 \text{ m}^4 = 10^{12} \text{ mm}^4$; unit slips here cost marks.

Q15 Answer: (A) 2 times the maximum bending stress and 8 times the maximum deflection

Concept: $M_{\max} = PL/4 \propto L$, so $\sigma \propto L$; $\delta = PL^3/(48EI) \propto L^3$.

Doubling L: $\sigma \times 2$, $\delta \times 8 \rightarrow \text{(A)}$.

Exam tip: Long spans are usually limited by deflection, short spans by strength.

What next?

Day 10 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124



Website test series



Dig Career Thrust app



Telegram: daily PDFs

Next, Day 11/90: Strength of Materials, Torsion, Thin Cylinders & Springs. Comment **GATE90** on our reel to get all links.

Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.